

Lecture 21

Goals

- Simon's Problem (in BQP)

→ Query Complexity: $M_Q \sim \sqrt{N}$

$$|\psi^n\rangle = \frac{1}{\sqrt{2}} |f(\vec{y})\rangle \otimes \frac{1}{\sqrt{2^n}} \sum_{\vec{z}} (-1)^{\vec{y} \cdot \vec{z}} \{1 + (-1)^{\vec{b} \cdot \vec{z}}\} |\vec{z}\rangle.$$

Dim of Hilbert space of upper register $\rightarrow 2^n$.

$\exists n$ linearly independent basis vectors
for the possible bit strings.

$\exists n-1$ L.I. non-zero $\vec{z}$ s.t. $\vec{b} \cdot \vec{z} = 0$
if $\vec{b} \neq 0$

Success prob. to find L.I. value of
 $\vec{z} \neq 0$ after finding $\vec{b}$

$$P_{k=1} = 1 - \frac{1}{2^n}$$

↑
probability: $\vec{z} = 0$.

$$P_{k=2} = 1 - \frac{1}{2^{n-1}}$$

$\swarrow$
 span generated by $\vec{v}_1, \vec{v}_2 \dots \vec{v}_l$ contains
 2^l vectors including $\vec{0}$

$$P_{l-1} = \frac{2^l}{2^{n-1}}$$



$$l = n-1$$

let query # be $n-1$ th $\swarrow$ # of failures
 $\rightarrow$ advance
 $m=0$ failures